Alessandro Linzi, PhD

Mathematician · Mathematics teacher · Researcher
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Appointments

ISIS Bonaldo Stringher (Udine) — Udine, Italy
Mathematics Teacher (class A026), full-time, 18 weekly hours — Sep 2026 – Present
Also taught at this school Nov 2025 – Jun 2026; current contract ends 30 Jun 2027.

ISIS Mattei Latisana — Latisana, Italy
Mathematics Teacher — Apr – Jun 2025

University of Nova Gorica — Nova Gorica, Slovenia
Assistant with PhD in the field of Mathematics — 2022 – 2024
Centre for Information Technologies and Applied Mathematics.

University of Szczecin — Szczecin, Poland
PhD Researcher and Teaching Assistant — 2018 – 2022
Supervised by Franz-Viktor Kuhlmann.

Universities of Florence and Udine — Florence / Udine, Italy
Teaching Assistant and Tutor — 2016 – 2018

Education and Qualifications

Università Telematica e-Campus
Italian secondary-school teaching habilitation (abilitazione), Class A026 — Mathematics — Jun 2026
Final exam 16 June 2026, grade 10/10; 60 CFU.

University of Szczecin — Szczecin, Poland
PhD in Mathematics (EU scholarship) — 2018 – 2022
Dissertation: *Algebraic hyperstructures in the model theory of valued fields.*

University of Florence — Florence, Italy
Master's in Mathematics, 110/110 cum laude — 2016 – 2018
University prize for exceptional performance.

University of Udine — Udine, Italy
Bachelor's in Mathematics — 2010 – 2015

Publications

2026 A. Linzi. A formal model of Cooperative Learning for Understanding and Epistemic Progress. *SN Social Sciences*. doi:10.1007/s43545-026-01807-9

2025 A. Linzi and P. Touchard. On the hyperfields associated to valued fields. *Journal of Pure and Applied Algebra*. doi:10.1016/j.jpaa.2025.107985

2025 A. Linzi. Notes on valuation theory for Krasner hyperfields. *Israel Journal of Mathematics*. doi:10.1007/s11856-025-2721-6

2025 A. Linzi and H. Stojalowska. Hypervaluations on hyperfields and ordered canonical hypergroups. *Iranian Journal of Mathematical Sciences and Informatics*.

2025 A. Linzi. On algebraically closed Krasner hyperfields. *Italian Journal of Pure and Applied Mathematics*.

2025 N. Cangiotti, A. Linzi and E. Talotti. L-mosaics and orthomodular lattices. *Italian Journal of Pure and Applied Mathematics*.

2025 A. Linzi and I. Cristea. Preface to the Special Issue "Algebraic Structures and Graph Theory, 2nd Edition". *Mathematics*. doi:10.3390/math13040577

2024 A. Linzi. Polygroup objects in regular categories. *AIMS Mathematics*. doi:10.3934/math.2024552

2023 A. Linzi. A result of Krasner in categorial form. *Mathematics*. doi:10.3390/math11244923

2023 A. Linzi and I. Cristea. Dependence relations and grade fuzzy set. *Symmetry*. doi:10.3390/sym15020311

2023 D. E. Kędzierski, A. Linzi and H. Stojalowska. Characteristic, C-characteristic and positive cones in hyperfields. *Mathematics*. doi:10.3390/math11030779

2022 K. Kuhlmann, A. Linzi and H. Stojalowska. Orderings and valuations in hyperfields. *Journal of Algebra*. doi:10.1016/j.jalgebra.2022.08.006

2019 P. Blaszkiewicz, H. Cmiel, A. Linzi and P. Szewczyk. Oettli-Théra and Caristi-Kirk ball spaces and applications. *Journal of Fixed Point Theory and Applications*. doi:10.1007/s11784-019-0729-4

Preprints and submissions. A. Linzi, *Valued Mosaics*, arXiv:2608.10616 (2026); A. Linzi, *Finite-field Krasner quotients: isomorphism thresholds, characteristics, and censuses*, arXiv:2608.03625 (2026); A. Linzi, *L-Mosaics and Bounded Join-Semilattices in Isabelle/HOL*, arXiv:2509.19854 (2025); A. Linzi, *A Horn Theory of Fields and Certain Hyperfields* (submitted, 2026).

Service

Editor, *Mathematics* (MDPI). Referee for international journals.

Skills and Languages

Teaching: secondary-school and university mathematics · **Mathematics:** model theory, valued fields, hypercompositional algebra, mathematical logic · **Tools:** LaTeX

Interests: AI ethics, strategy, and industry; the theory of AI

Languages: Italian (native), English (fluent), Spanish (fluent), Slovenian (basic), Polish (basic)